Week 4

Linear Regression

$$y = \beta_0 + \beta_1 x$$

Residual Sum of Squares

Fitting Loss Function

Given data $\{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}, x_i \in \mathbb{R}^d, y_i \in \mathbb{R}, i = 1, 2, \dots, n$

$$L(\theta) = \frac{1}{2}\sum_{i=1}^{n}(x_i^T\theta - y_i)^2$$

Minimising L

Feature Matrix A

$$A = \begin{bmatrix} x_1^T \\ x_2^T \\ . \\ . \\ . \\ x_n^T \end{bmatrix} Y = \begin{bmatrix} y_1 \\ y_2 \\ . \\ . \\ . \\ y_n \end{bmatrix} A\theta = \begin{bmatrix} x_1^T\theta \\ x_2^T\theta \\ . \\ . \\ . \\ x_n^T\theta \end{bmatrix} A\theta - Y = \begin{bmatrix} x_1^T\theta - y_1 \\ x_2^T\theta - y_2 \\ . \\ . \\ . \\ x_n^T\theta - y_n \end{bmatrix}$$

$$(A\theta - Y)^T(A\theta - Y) = \sum_{i=1}^{n}(x_i^T\theta - y_i)^2$$

So,

$$L(\theta) = \frac{1}{2}(A\theta - Y)^T(A\theta - Y)$$

Least Squares Solution

$$(A^TA)\theta = A^TY$$

> 💡 Least square regression solves a maximum likelihood estimation problem under a linear model.

Can we write $\theta = (A^TA)^{-1}A^TY$?

Yes, if A is full rank.

Eigenvalues and Eigenvectors

Let $A \in \mathbb{R}^n$, we say that a non-zero vector $v \in \mathbb{R}^n$ is an eigenvector of A corresponding to eigenvalue λ (a scalar) if

$$Av = \lambda v$$

The above relation tells that when the matrix (linear transformation) A applied on v, then it scales v by an amount λ.

How to find?

Step 1: Put Identity matrix I in the above equation.

$$Av = \lambda Iv$$

Step 2: Bring all to left side

$$Av - \lambda Iv = 0$$

Step 3: If v is non-zero, then we can solve for λ using the determinant.

$$|A - \lambda I| = 0$$

Trick for Characteristic Equation for 2×2 matrix

$$\lambda^2 - (\text{Trace of A})\lambda + \det A$$

Trick for Characteristic Equation for 3×3 matrix

$$\lambda^3 - (\text{Trace of A})\lambda^2 + (\text{Sum of minors along diagonal})\lambda - \det A$$

Properties

- When A is squared, the eigenvectors stay the same. The eigenvalues are squared.
- The reflection matrix $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues 1 and -1.
- When a matrix is shifted by I , each λ is shifted by 1. No change in eigenvectors.
- The product of the n eigenvalues equals the determinant.
- The sum of the n eigenvalues equals the sum of the n diagonal entries.
- If an eigenvalue of a matrix A is zero, then its corresponding eigenvector belongs to the null space of A.
- For a real symmetric matrix, the eigenvalues are always real.
- For a matrix A with eigenvalue λ, the corresponding eigenvector lies in the null space of $A - \lambda I$.
- If $A = aI + B$ where A and B are matrices and a is a scalar, eigenvectors of A are the same as the eigenvectors of B.
- For a matrix A having eigenvalue λ, $A - \lambda I$ is a singular matrix (det is 0).
- If x is an eigenvector of A corresponding to eigenvalue λ_1 and x is also an eigenvector of B corresponding to eigenvalue λ_2 , then x is also an eigenvector of $(A + B)$.
- A permutation matrix has eigenvalues 1 or -1.
- If P is a projection matrix, then the eigenvalue corresponding to every non-zero vector lying in the column space of P is 1.

Diagonalization of Matrix

- If all eigenvalues of a matrix are distinct, the matrix is diagonalizable.
- If all eigenvectors of a matrix are linearly independent, then the matrix is diagonalizable.
- A matrix is diagonalizable if algebraic and geometric multiplicity of eigenvalues are same.

Fibonacci Sequence using Diagonalisation

The n^{th} term of the Fibonacci sequence is approximately given by

$$\frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n$$